

Test 2 (PHYS 235, Calculus track) — Full Solutions

Constants used throughout: $\frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$, $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/(\text{N}\cdot\text{m}^2)$, $e = 1.602 \times 10^{-19} \text{ C}$, $m_p = 1.673 \times 10^{-27} \text{ kg}$.

Section 1 — P1: sign of V , direction of $\vec{E}$ (4 pts)

$Q = +4.0 \text{ nC}$ sits at the origin, $q = -1.0 \text{ nC}$ sits at $x = 5.0 \text{ cm}$. P1 is on the axis at $x = 4.0 \text{ cm}$, 4.0 cm from Q and 1.0 cm from q .

SOLVE: Superpose the two point-charge potentials (dropping the common factor $1/4\pi\epsilon_0$ and units, as the hint allows):

$$V_1 \propto \frac{Q}{r_Q} + \frac{q}{r_q} = \frac{4.0}{4.0} + \frac{-1.0}{1.0} = 1.0 - 1.0 = \boxed{0}.$$

(C) zero. For the field: Q is positive and P1 is to its right, so $\vec{E}_Q$ points away from Q , in $+\hat{x}$. q is negative and P1 is to its left, so $\vec{E}_q$ points *toward* q – also $+\hat{x}$. The two contributions add instead of fighting each other, so

$$\boxed{\vec{E}_1 \text{ points in } +\hat{x} \text{ (due right, toward } q\text{)}}.$$

REVIEW: P1 is the case where $V = 0$ but $\vec{E} \neq 0$ – the two potentials cancel exactly (closer to the smaller-magnitude charge by exactly the ratio that makes it work), but the two field contributions don't fight each other at all, they add. $V = 0$ and $\vec{E} = 0$ are not the same condition, and this point is designed to make that visible.

Section 1 — P2: sign of V , direction of $\vec{E}$ (4 pts)

P2 is at (5.0, 2.4) cm – directly above q , 2.4 cm away from it, and $\sqrt{5.0^2 + 2.4^2} = 5.5$ cm from Q .

SOLVE:

$$V_2 \propto \frac{4.0}{5.5} + \frac{-1.0}{2.4} = 0.73 - 0.42 = \boxed{+0.30} \Rightarrow \textbf{(A) positive}.$$

For the field, weight each unit vector by $1/r^2$. $\vec{E}_Q$ points away from Q , along the line from Q to P2 (up and to the right); $\vec{E}_q$ points straight down toward q (P2 sits directly above it). Using $E \propto |charge|/r^2$:

$$\vec{E}_{Q,2} \propto \frac{4.0}{5.5^2} (0.90, 0.43) = (0.117, 0.056), \quad \vec{E}_{q,2} \propto \frac{1.0}{2.4^2} (0, -1) = (0, -0.174).$$

Adding, $\vec{E}_2 \propto (0.117, -0.117)$ – equal magnitude $+x$ and $-y$ components, i.e.

$$\boxed{\vec{E}_2 \text{ points } 45^\circ \text{ below } +\hat{x} \text{ (southeast)}}.$$

REVIEW: The 2.4 cm height isn't an arbitrary number – it's exactly the value that makes Q 's field and q 's field add up to a clean 45° direction, so a numeric sanity check (rather than just a qualitative “down-and-to-the-right” guess) is the fastest way to confirm you have the right compass direction.

Section 1 — P3: sign of V , direction of $\vec{E}$ (4 pts)

P3 is on the axis at $x = 10$ cm, beyond q : 10 cm from Q , 5.0 cm from q .

SOLVE:

$$V_3 \propto \frac{4.0}{10} + \frac{-1.0}{5.0} = 0.40 - 0.20 = \boxed{+0.20} \Rightarrow \textbf{(A) positive}.$$

For the field: $\vec{E}_Q \propto Q/r_Q^2 = 4.0/10^2 = 0.040$, pointing away from Q (i.e. $+\hat{x}$, since P3 is to the right of Q). $\vec{E}_q \propto |q|/r_q^2 = 1.0/5.0^2 = 0.040$, pointing *toward* q (i.e. $-\hat{x}$, since P3 is to the right of q). Equal magnitude, opposite direction:

$$\boxed{\vec{E}_3 = 0}.$$

REVIEW: P3 is the mirror image of P1: there, V canceled but $\vec{E}$ didn't; here $\vec{E}$ cancels exactly (by design: $4.0/10^2$ and $1.0/5.0^2$ both equal 0.040) while V stays comfortably positive. Together the three points cover all three cases a student needs to tell apart: $V = 0, \vec{E} \neq 0$ (P1); both nonzero (P2); $V \neq 0, \vec{E} = 0$ (P3).

Section 2 — Part 1: the dipole moment (5 pts)

Charges of magnitude 5 nC, separated by 6 μm , with $\vec{E} = 20 \text{ N/C } \hat{y}$. The dipole moment points 60° below the x -axis (“Orientation A”).

SOLVE: The dipole moment has magnitude

$$p = qd = (5.0 \times 10^{-9} \text{ C})(6.0 \times 10^{-6} \text{ m}) = \boxed{3.0 \times 10^{-14} \text{ C} \cdot \text{m}},$$

pointing (by definition, from $-q$ to $+q$) 60° below the $+x$ -axis, i.e.

$$\vec{p} = p(\cos(-60^\circ), \sin(-60^\circ), 0) = (1.5, -2.6, 0) \times 10^{-14} \text{ C} \cdot \text{m}.$$

Drawing: the $+x$ -axis horizontal, $+y$ vertical. $\vec{E}$ is a vertical arrow pointing straight up ($+\hat{y}$). $\vec{p}$ is a separate arrow starting at the dipole’s center, 60° *below* the $+x$ -axis (down and to the right, into the fourth quadrant) – *not* aligned with $\vec{E}$ and not symmetric about either axis. The negative charge sits at the tail of $\vec{p}$, the positive charge at the head, 6 μm apart along that same line.

REVIEW: A common error here is drawing p at 60° *above* the axis, or measuring the angle from $\vec{E}$ instead of from the x -axis – read the setup’s wording carefully (as the problem itself warns): the angle given is always relative to the axis stated, not to the field.

Section 2 — Part 2: torque (4 pts)

Same dipole and field as above.

SOLVE: $\vec{\tau} = \vec{p} \times \vec{E}$. Since $\vec{p} = (p_x, p_y, 0)$ and $\vec{E} = (0, E_y, 0)$ (purely $\hat{y}$),

$$\vec{\tau} = (0, 0, p_x E_y) = (0, 0, (1.5 \times 10^{-14})(20)) = \boxed{(0, 0, 3.0 \times 10^{-13}) \text{ N} \cdot \text{m}},$$

i.e. $3.0 \times 10^{-13} \text{ N} \cdot \text{m}$ in $+\hat{z}$ (out of the page).

REVIEW: The torque vector's direction is the rotation *axis*, found by the right-hand rule, not a push in some in-plane direction. $+\hat{z}$ means: curl the right hand's fingers from $\vec{p}$ toward $\vec{E}$ through the *smaller* angle between them, and the thumb points out of the page – confirming the dipole is being spun counterclockwise (as drawn, with $+x$ right and $+y$ up), rotating $\vec{p}$ up and to the left, toward alignment with $\vec{E}$.

Section 2 — Part 3: potential energy, Orientation A vs. B (6 pts)

Same dipole and field. Orientation B is whichever orientation has the lowest PE.

SOLVE: $U = -\vec{p} \cdot \vec{E} = -pE \cos \theta$, where θ is the angle between $\vec{p}$ and $\vec{E}$. This is minimized when $\cos \theta = 1$, i.e. $\vec{p}$ points the *same direction as* $\vec{E}$ – straight up, $+\hat{y}$. **Orientation B: $\vec{p}$ parallel to $\vec{E}$ (pointing straight up).**

In Orientation A, $\vec{p}$ is at -60° and $\vec{E}$ is at $+90^\circ$ (both measured from $+x$), so $\theta_A = 90^\circ - (-60^\circ) = 150^\circ$:

$$U_A = -pE \cos 150^\circ = -(3.0 \times 10^{-14})(20)(-0.866) = +5.2 \times 10^{-13} \text{ J}.$$

In Orientation B, $\theta_B = 0$:

$$U_B = -pE \cos 0^\circ = -(3.0 \times 10^{-14})(20) = -6.0 \times 10^{-13} \text{ J}.$$

So Orientation A has

$$U_A - U_B = 5.2 \times 10^{-13} - (-6.0 \times 10^{-13}) = \boxed{1.1 \times 10^{-12} \text{ J more than Orientation B}}.$$

REVIEW: $U_A - U_B = pE(1 - \cos \theta_A)$ ranges from 0 (if A already equals B) to $2pE = 1.2 \times 10^{-12} \text{ J}$ (if A is the *worst* orientation, anti-aligned with $\vec{E}$). 150° is close to that worst case (180°), so it makes sense that $U_A - U_B$ comes out close to, but just under, that $2pE$ ceiling.

Section 3 — Part 1: field at A, B, C (10 pts)

Positive rod (λ_0) and negative rod ($-4\lambda_0$) separated by $2d$. A is a distance d to the left of the positive rod; C is a distance d to the right of the negative rod; B is halfway between (distance d from each).

SOLVE: Put the positive rod at $x = 0$ and the negative rod at $x = 2d$, so $A = -d$, $B = d$, $C = 3d$. All three points lie on the line through both rods, so each rod's field there is purely along $\hat{x}$: pointing *away* from the positive rod, *toward* the negative rod.

At A ($r_{\text{pos}} = d$, $r_{\text{neg}} = 3d$): the positive rod's field pushes left ($-\hat{x}$, away from it); the negative rod's field pulls right ($+\hat{x}$, toward it, since the negative rod is to A's right).

$$E_A = -\frac{\lambda_0}{2\pi\epsilon_0 d} + \frac{4\lambda_0}{2\pi\epsilon_0(3d)} = \frac{\lambda_0}{2\pi\epsilon_0 d} \left(-1 + \frac{4}{3} \right) = \boxed{\frac{\lambda_0}{6\pi\epsilon_0 d} \quad (+\hat{x}, \text{toward the rods})}.$$

At B ($r_{\text{pos}} = r_{\text{neg}} = d$): now *both* fields point the same way – away from the (weaker) positive rod is $+\hat{x}$ here (B is to its right), and toward the (stronger) negative rod is also $+\hat{x}$ (B is to its left). They add, not cancel:

$$E_B = \frac{\lambda_0}{2\pi\epsilon_0 d} + \frac{4\lambda_0}{2\pi\epsilon_0 d} = \boxed{\frac{5\lambda_0}{2\pi\epsilon_0 d} \quad (+\hat{x})}.$$

At C ($r_{\text{pos}} = 3d$, $r_{\text{neg}} = d$): away from the positive rod is $+\hat{x}$ (C is to its right); toward the negative rod is $-\hat{x}$ (C is to its right of it, so “toward” means back to the left).

$$E_C = \frac{\lambda_0}{2\pi\epsilon_0(3d)} - \frac{4\lambda_0}{2\pi\epsilon_0 d} = \frac{\lambda_0}{2\pi\epsilon_0 d} \left(\frac{1}{3} - 4 \right) = \boxed{-\frac{11\lambda_0}{6\pi\epsilon_0 d} \quad (\text{i.e. } \frac{11\lambda_0}{6\pi\epsilon_0 d} \text{ in } -\hat{x}, \text{toward the rods})}.$$

With $d = 6$ cm and $\lambda_0 = 400$ nC/m (the 80%-credit numeric option): $E_A \approx 4.0 \times 10^4$ N/C, $E_B \approx 6.0 \times 10^5$ N/C, $E_C \approx 4.4 \times 10^5$ N/C.

REVIEW: B and C both end up dominated by the much closer, 4×-stronger negative rod (that's why both exceed E_A by more than an order of magnitude), but A is the one case where the two contributions subtract instead of reinforcing – which is exactly why A's field comes out so much smaller than the others, even though it's also only a distance d from its near rod, same as B and C.

Section 3 — Part 2: is there a location where $\vec{E} = 0$? (6 pts)

Same two rods, now fixing $d = 6$ cm.

SOLVE: First: *where* could a null point even be possible? At any point off the line through the two rods, the two rods' fields point in genuinely different (non-opposite) directions – two nonzero vectors that aren't anti-parallel can never sum to zero, regardless of their magnitudes. So a null point, if one exists, must lie on the line through both rods (the same line A , B , C sit on).

On that line there are three regions to check. *Between the rods* ($0 < x < 2d$): both fields point the same way (as at B above) – they can only add, never cancel, so no zero there. *Right of the negative rod* ($x > 2d$, like C): setting the two magnitudes equal, $\lambda_0/r_{\text{pos}} = 4\lambda_0/r_{\text{neg}}$ with $r_{\text{neg}} = r_{\text{pos}} - 2d$, gives $x = -2d/3$ – negative, so *not* actually in this region. No zero here either.

Left of the positive rod ($x < 0$, like A): here the positive rod's field points further left ($-\hat{x}$) and the negative rod's field points right ($+\hat{x}$, toward it) – opposite directions, so cancellation is at least possible. Setting magnitudes equal, with $r_{\text{pos}} = |x| = -x$ and $r_{\text{neg}} = 2d - x$:

$$\frac{\lambda_0}{-x} = \frac{4\lambda_0}{2d - x} \Rightarrow 2d - x = -4x \Rightarrow \boxed{x = -\frac{2d}{3}, \text{ i.e. a distance } \frac{2}{3}d \text{ to the left of the positive rod}}.$$

With $d = 6$ cm, that's 4.0 cm to the left of the positive (left) rod – between the positive rod and point A (which sits a full $d = 6$ cm out).

REVIEW: It makes sense the null point sits *closer* to the positive rod than A does: since the negative rod is $4\times$ stronger, the field point has to huddle in close to the weaker rod – well inside A 's distance – for the weaker rod's $1/r$ advantage (from proximity) to offset the stronger rod's $4\times$ charge advantage.

Section 4: Gauss's Law

Part 1: enclosed charge (8 pts)

A cube with 3.0 cm sides. The field is purely in $\hat{y}$ everywhere. At the top face, $\vec{E} = \langle 0, -34, 0 \rangle$ N/C; at the bottom face, $\vec{E} = \langle 0, +20, 0 \rangle$ N/C.

SOLVE: Since $\vec{E}$ has no x or z component, the flux through all four side faces is zero (their outward normals are $\pm\hat{x}$ or $\pm\hat{z}$, perpendicular to $\vec{E}$). Only the top and bottom faces contribute. With $A = (0.030 \text{ m})^2 = 9.0 \times 10^{-4} \text{ m}^2$:

Top face, outward normal $+\hat{y}$: $\Phi_{\text{top}} = \vec{E}_{\text{top}} \cdot \hat{n} A = (-34)(+1)A = -34A$.

Bottom face, outward normal $-\hat{y}$: $\Phi_{\text{bottom}} = \vec{E}_{\text{bottom}} \cdot \hat{n} A = (+20)(-1)A = -20A$.

$$\Phi_{\text{total}} = -34A - 20A = -54A = -54(9.0 \times 10^{-4}) = -0.0486 \text{ N} \cdot \text{m}^2/\text{C}.$$

By Gauss's law,

$$Q_{\text{enc}} = \varepsilon_0 \Phi_{\text{total}} = (8.85 \times 10^{-12})(-0.0486) = \boxed{-4.3 \times 10^{-13} \text{ C} \approx -0.43 \text{ pC}}.$$

A drawing should show the cube with $\vec{E}$ arrows pointing *down* into the top face (since $\vec{E}_{\text{top}}$ points in $-\hat{y}$, into the cube from above) and pointing *up* into the bottom face (since $\vec{E}_{\text{bottom}}$ points in $+\hat{y}$, into the cube from below) – field lines converging into the cube from both the top and the bottom, consistent with a net negative charge trapped inside.

REVIEW: The two faces don't need to have equal field strength for this to work – Gauss's law only cares about the net flux. Here 34 and 20 N/C aren't equal, so the two faces don't exactly cancel, and that imbalance (both pointing inward, but by different amounts) is exactly what leaves a nonzero, negative Q_{enc} .

Part 2: what Gauss's law can and can't tell you (4 pts)

Same cube.

SOLVE: Gauss's law only ever gives the *total* charge enclosed by a surface – it has no way to say *where* inside the cube that charge actually sits. The -0.43 pC found above could be a thin sheet right at the cube's center, spread uniformly through the whole volume, concentrated near one face, or any other distribution; this calculation can't distinguish between them, and nothing in the given field values (just two numbers, at the top and bottom) provides enough information to localize it further.

For charge *outside* the cube: this particular Gaussian surface (the cube itself) tells us nothing about it. To learn anything about charge outside, you'd need to draw a *different* Gaussian surface that encloses some of that outside region, together with field information on that new surface – neither of which this problem gives us. The one thing we *can* say is that whatever is outside is consistent with the field being purely $\hat{y}$ -directed everywhere (not just inside the cube): that's a strong symmetry statement (reminiscent of infinite charged sheets, where the field depends only on height, never on the horizontal position), but it still doesn't pin down a location or an amount.

REVIEW: This pair of questions is really testing the same idea from two sides: Gauss's law is a statement about a *surface integral* of the field, which only ever recovers the enclosed charge's *total*, not its spatial distribution, and only for the one surface you chose to apply it to.